MILITARY SPECIFICATION

PERISCOPE, BATTERY COMMAND: M65

1. SCOPE

1.1 This specification covers one type of optical mechanical instrument known as Periscope Battery Command: M65.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

US Army, Frankford Arsenal

F7579932	Periscope, Battery Command: M65
F8565556	Pressure Tester, Two Station

FSC 6650

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PACKAGING DATA SHEET

8268671 Packaging of Periscope, Battery
Command: M65

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The periscope shall be manufactured in accordance with Drawing F7579932.

3.2 General specification.- The contractor shall be responsible for compliance with the following requirements of MIL-F-13926:

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking

3.3 Temperature conditions.- The requirements of this specification shall be met with the periscope thermally stabilized at an ambient temperature of between 60° and 90° Fahrenheit (F) unless otherwise specified in the requirements paragraphs.

3.4 Workmanship.- The workmanship requirements of MIL-F-13926 shall be met. The exterior protective surface finishes shall be free of any defects which effect their protective qualities and appearances. Optical surfaces shall be clean and free of any scratches or blemishes which may effect their optical qualities or appearance.

3.5 Environmental conditions.- Any periscope subjected to any or all of the requirements of 3.5.1 to 3.7 inclusive shall subsequently meet the requirements of 3.8 to 3.9.21 inclusive, unless otherwise specified.

3.5.1 Storage temperature.- The periscope shall show no evidence of physical failure such as cement separation, glass breakage or casting failure, after having been exposed and thermally stabilized at ambient temperatures of -80° and +160° Fahrenheit.

3.5.2 Extreme operating temperatures.- The periscope shall meet the requirements of 3.9.21 while exposed and thermally stabilized at ambient temperatures of -40° and +150°F.

3.6 Vibration.- The periscope shall be positioned with the left and right periscope housings secured in a horizontal plane. The periscope shall be vibrated at a constant frequency of 30 cycles per second with a single amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of five minutes plus or minus 15 seconds at room temperature, 60° to 90°F.

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3.7 Sealing and flushings.- The left and right periscopes shall each be pressurized to an internal pressure of 5 psig, using dry nitrogen having a dew point at least as low as -25°F. The pressure on each shall be maintained at 5 psig for a period of five minutes to balance pressure within the instruments internal cells. Subsequent to the five minutes balancing period the periscope shall show no evidence of leakage with an internal pressure of 5 psig over a period of 5 minutes ± 15 seconds. The internal pressure of each periscope shall then be reduced to zero psig. Each periscope shall then be flushed with dry nitrogen until the interior atmosphere of each periscope has a dew point at least as low as -25°F. Each periscope shall then be sealed with an internal pressure of between 0.25 and 0.50 psig of dry nitrogen.

3.8 Cleanliness.- The periscope shall meet the cleanliness requirements of MIL-O-13830. There shall be no more than one particle of dirt visible on the surface of the reticle, the area of the particles shall not be in excess of the area equivalent to a size 10 dig as indicated in the focal plane of the right eyepiece. There shall be no dirt on other optical surfaces of either periscope which interferes with observation or adversely effects image quality.

3.9 Accuracy.

3.9.1 Orientation.- With the periscope mounted on a vertical spindle, the angle of site index shall be in coincidence with the number "3" on the angle of site scale and the micrometer scale at zero. The level vial shall be centered within the thickness of a vial graduation line width. From this initial position the requirements of 3.9.2 to 3.9.19 shall be met.

3.9.2 Level line.- The line of sight through the geometric axis of the reticle shall be in coincidence with an incoming horizontal line of sight within 0.50 mil in elevation.

3.9.3 Collimation.- Parallel rays entering the objectives shall emerge from the eyepiece parallel within 15 minutes of arc in dipvergence and shall not diverge less than 5 nor more than 30 minutes of arc. These conditions shall be met at infinity focus setting of the eyepieces, and for any setting of the interpupillary scale between 60 and 72 millimeters.

3.9.4 Resolution.- Target resolution of each periscope on the optical axis shall be 7 seconds of arc or less with each filter knob at the clear position. Resolution consists of detectable line structure and proper line count in all four meridians. Focus variation between any two meridians shall not exceed 0.25 diopters at the eyepiece.

3.9.5 Parallax.- Parallax between the target image and the center of the periscope reticle shall not exceed 0.10 mil with a target at 1000 ± 100 meters.

3.9.6 Magnification.- Magnification of the right and left optical systems shall not differ from each other by more than two percent.

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3.9.7 Filters.- There shall be complete filter coverage for each periscope. The filter knobs shall indicate the correct filter position. There shall be no overlap of adjacent filters in the field of view.

3.9.8 Reticle accuracy.- The angular separation between the left and right 30 mil graduation on the horizontal reticle line through the geometric axis of the reticle shall be 60 mils ± 2.0 mils.

3.9.9 Image tilt.- The image of a plumb line formed by the two optical systems shall be vertical within one degree and parallel to each other within 0.5 degree.

3.9.10 Parallelism of reticle and image.- The vertical reticle line through the geometric axis of the reticle, shall be parallel to the image of a plumb line within ± 30 minutes.

3.9.11 Eyepiece focus.- Eyepiece focus shall be adjustable over the range of at least plus and minus 3 diopters as read on each eyepiece scale. The zero diopter setting of each eyepiece scale shall indicate zero diopter within ± 0.5 diopter when infinity target images are brought into best focus.

3.9.12 Reticle focus.- The periscope reticle shall be in sharp focus at zero diopter within ± 0.5 diopter.

3.9.13 Interpupillary adjustment.- The eyepiece shall be adjustable between 60 and 72 millimeters. The interpupillary scale shall indicate the correct interpupillary distance at 66 millimeters.

3.9.14 Elevation excursion.- Rotation of the elevation knob shall elevate the periscope from a position at least 320 mils below a horizontal line of sight to a position at least 320 mils above a horizontal line of sight.

3.9.15 Chatter in elevation or depression movement.- Verical movement (jumps) in elevation and depression movement shall not exceed 2.0 mils at any point over the excursion of 640 mils in each direction.

3.9.16 Plumb travel.- The line of sight shall be in coincidence with a plumb line target within 1.0 mil over an excursion of 640 mils in elevation, travel starting at minus 320 mils depression below a horizontal line of sight and elevating to plus 320 mils above a horizontal line of sight. In addition the total spread between any two error readings shall not exceed 1.0 mil.

3.9.17 Azimuth displacement.- The azimuth shift in the line of sight (at horizontal position, Index 3, micrometer 0) shall not exceed 0.2 mil after a change of 300 mils in elevation and 300 mils in depression from a horizontal position, with an angular velocity of 20.0 mils per second or less.

3.9.18 Angle of site error.- The error of any angle of site reading shall not exceed 1.0 mil excluding backlash.

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3.9.19 Backlash angle of site.- Backlash in the angle of site mechanism shall not exceed 1.0 mil.

3.9.20 Illumination.- With illumination at the red window adjacent to the reticle as provided by a Lamp MS51608-3 (Commercial Identification #325) energized from a 3 volt source, the reticle pattern shall be clearly defined when the cap is positioned to cover the objective window of the right periscope.

3.9.21 Operability.- The periscope shall meet the applicable operability requirements of Table I when checked at the temperatures indicated therein. Torque values are maximum for breaking torque, (maximum torque to start motion) and rotation to maximum clockwise and counterclockwise limits of motion.

TABLE I

	-40°F	60 to 90°F	+150°F
Angle of site knob	6 in. lb.	3 in. lb.	3 in. lb.
Elevation knob	13 in. lb.	10 in. lb.	10 in. lb.
Diopter adjustment (left) (right)	8 in. lb.	5 in. lb.	5 in. lb.
Interpupillary knob	6 in. lb.	3 in. lb.	3 in. lb.
Filter selectors	5 in. lb.	2 in. lb.	2 in. lb.
Release plunger	operable	operable	operable

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) periscopes plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and proccedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

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4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables II, III and IV herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables II, III and IV shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE II - CLASSIFICATION OF DEFECTS

CLASS		REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE			
MAJOR: AQL 1.0% DEFECTIVE			
101.	Vibration	3.6	4.6.3
102.	Sealing & flushing	3.7	4.6.4
103.	Cleanliness	3.8	4.6.5
104.	Resolution	3.9.4	4.6.6
105.	Scales and level vial	3.9.1	4.6.7.1
106.	Level line	3.9.2	4.6.7.2
107.	Collimation	3.9.3	4.6.7.3
108.	Parallax	3.9.5	4.6.7.4
109.	Magnification	3.9.6	4.6.7.5
110.	Filters	3.9.7	4.6.7.6
111.	Reticle accuracy	3.9.8	4.6.7.7

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TABLE II - CLASSIFICATION OF DEFECTS (Cont'd)

		REQUIREMENT	TEST PROCEDURE
112.	Image tilt	3.9.9	4.6.7.8
113.	Parallelism of reticle & image	3.9.10	4.6.7.9
114.	Eyepiece focus	3.9.11	4.6.7.10
115.	Reticle focus	3.9.12	4.6.7.11
116.	Interpupillary adjustment	3.9.13	4.6.7.12
117.	Azimuth displacement	3.9.17	4.6.7.13
118.	Elevation excursion	3.9.14	4.6.8.1
119.	Chatter in elevation or depression movement	3.9.15	4.6.8.2
120.	Plumb travel	3.9.16	4.6.8.3
121.	Angle of site error	3.9.18	4.6.8.4
122.	Backlash angle of site	3.9.19	4.6.8.5
123.	Illumination	3.9.20	4.6.9
124.	Operability	3.9.21	4.6.10
125.	Workmanship	3.4	4.6.11

MINOR: NONE

NOTE: The tests in Table II shall be conducted at a temperature between +60° and +90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One periscope shall be selected at random as a special sample from each 100 produced. The sample shall meet the requirements and tests in Table III.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable Draw- Visual
302.	General specification	3.2	MIL-F-13926 - Visual

NOTE: The tests in Table III shall be conducted at a temperature between +60° and +90°F.

4.4.2 Environmental.

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4.4.2.1 Periscope.- Three periscopes shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall have met the requirements and tests in Table IV and shall then meet the requirements and tests in Table II.

TABLE IV

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Storage temperature (-80° and +160°F)	3.5.1	4.6.1
304.	Extreme operating temperatures (Torque -40° and +150°F)	3.5.2, 3.9.21	4.6.2, 4.6.10
305.	Extreme operating temperatures (Functional operability -40° and +150°F)	3.5.2, 3.9.21	4.6.2, 4.6.10

4.4.3 Failure of sample.- Should any item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment

4.5.2.1 Government design.- All inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the technical data package list.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

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4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Orientation.- Special testing equipment to test the periscope assembly shall conform to the following requirements:

(a) Periscope mounting spindle. The periscope mounting spindle shall meet the requirements set forth in Figure 2.

(b) Leveling of special testing equipment with periscope mounting spindle. The qualified surface of the testing equipment utilized for securing the spindle and target projector collimators shall be cross leveled within 10 seconds. Provisions shall be made on the testing equipment for locating and securing the mounting spindle (Figure 2) utilized to assure accurate and repeatability positioning of the periscope assembly. A precision level shall be used to align the axis of the mounting spindle vertically within 10 seconds.

(c) Target collimators. The two target project collimators shall have a clear aperture of at least 2.0 inches and contain appropriate reticle patterns for measuring the mechanical and optical accuracies of the periscope assembly. The target collimators shall be set at infinity focus, unless otherwise specified herein. The zero lines of sight of the target collimators shall be set with the lines of sight in the same horizontal (level) plane (approximately 12.25 inches above the nominal line of sight of the periscope assembly) within 5 seconds. The lines of sight of the target collimators shall be parallel to each other.

(d) Viewing double collimator. The emergent rays of the periscope shall be measured with the aid of a double collimator aligned to the geometric optical axis of the periscope eyepiece assemblies. The viewing collimators shall contain appropriate reticle patterns for measuring collimation of the periscope. The viewing collimator lines of sight shall be parallel and lay in the same horizontal plane. The vertical reticle lines of the collimators shall be aligned plumb when in the test position.

4.5.2.3.2 Performance - periscope assembly.- Special testing equipment to test the periscope assembly shall conform to the following requirements:

(a) Periscope mounting spindle.- The periscope mounting spindle shall meet the requirements in Figure 2.

(b) Target alinement adapter.- The target alinement adapter shall conform to the requirements in Figure 3 and pre-adjusted prior to performing any tests as follows:

(1) Assemble adapter to periscope mounting spindle (see Figure 1) assuring that the key slot of the adapter is positioned on the key of the spindle. With the base of the spindle on the leveled and cross leveled surface, adjust the line of sight of the periscope parallel to the base of the spindle within 10 seconds and lock the periscope in position.

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(c) Elevation test fixture.- The elevation test fixture (see Figure 1) shall be provided with an elevation scale and vernier, periscope mounting spindle and target alinement adapter as outlined below:

(1) The elevation scale and vernier shall be graduated and calibrated to read required angles to an accuracy of .1 mil.

(2) The periscope mounting spindle (see Figure 2) shall be secured to the elevation spindle. The axis of the periscope spindle shall be perpendicular to and intersect the elevation spindle axis within 20 seconds. The longitudinal axis of the key shall be parallel to the elevation spindle axis within 10 seconds.

(3) With the target alinement adapter secured to the periscope mounting spindle and the elevation spindle axis in a horizontal (leveled) plane, the periscope line of sight shall track a plumb line to an accuracy of .1 mil.

(d) Wall targets.- A calibrated wall target (see Figure 1) capable of measuring the optical and mechanical accuracies of the periscope assembly shall conform to the following requirements:

(1) The "0" (horizontal line of sight) of the target image shall be positioned 12.25 inches above the geometric optical axis of the eyepiece assembly. The depression and elevation target images shall be positioned in the same vertical plane with the "0" (horizontal line of sight). The angular position of the lines of sight of the depression and elevation target images with tolerance mil blocks shall be established at the following depressions and elevations in mils, -320, -300, -200, -100, +100, +200, +300 and +320 mils to an accuracy of .1 of a mil.

(2) A calibrated vertical image line (plumb line) for measuring plumb travel accuracy of the elevation test fixture prior to performing any accuracy tests of the periscope assembly shall be incorporated on the wall target.

4.5.2.3.3 Sealing periscope.- Special testing equipment to test sealing of the periscope assembly shall conform to the internal pressure test requirements of Drawing F8565556, Pressure Tester-Two Station.

4.6 Test methods and procedures.

4.6.1 Storage temperature.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and conditions of 3.5.1. The periscope assembly shall be exposed to each temperature extreme specified in 3.5.1 for a minimum of 5 hours. After the temperature exposures, the periscope assembly shall be stabilized at an ambient temperature between +60° and +90° Fahrenheit and subjected to the appropriate visual and tactile examination of 4.6.3.1. The periscope shall conform to the requirements of 3.5.1. Caution shall be exercised during this test to avoid subjecting the periscope to thermal shock.

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4.6.2 Extreme operating temperatures.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and conditions of 3.5.2. The periscope assembly shall be exposed to each temperature extreme specified in 3.5.2 for a period of 5 hours. At the end of each 5 hour period, and while still at the temperature extreme, the periscope shall be subjected to the applicable test of 4.6.10, to determine conformance to the requirements of 3.9.21 at the temperature indicated. Caution shall be exercised during this test to avoid subjecting the periscope to thermal shock.

4.6.3 Vibration.- Standard vibrating equipment capable of providing the amplitude and frequency as specified in 3.6 shall be utilized for the vibration test. Unless otherwise specified, the test equipment shall be in 3 planes in accordance with MIL-F-13926. The periscope assembly shall be vibrated in the direction and conditions specified in 3.6. Upon completion of the vibration test the periscope assembly shall be subjected to the visual and tactile examination specified in 4.6.3.1.

4.6.3.1 Examination.- Subsequent to vibration, the periscope shall be thoroughly examined for physical defects prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts. A tactile examination shall also be conducted to determine looseness or improper functioning of moving parts such as diopter movement, elevation, micrometer knobs and release latch. An examination shall also be made for loose internal parts or broken optical components, such as (lenses, prisms, head prisms and reticle). Any one of the possible defects defined above shall be cause to consider a periscope defective. Periscope assemblies which have passed the above examinations shall be subjected to the tests specified in 4.6.4 through 4.6.11 inclusive.

4.6.4 Sealing and flushing.- Internal pressurizing of the periscope to test for sealing shall be performed using nitrogen gas having a dew point at least as low as -25°F, and special testing equipment conforming to 4.5.2.3.3. The periscope shall be pressurized to the requirements of 3.7 in accordance with the procedure for internal pressure testing of Drawing F8565556. Maintain the applied pressure for the period specified to balance pressure within the periscope cells. With the pressure at 5 psig the periscope shall retain this pressure for 5 minutes $\pm$ 15 seconds with no indication of leakage. The pressure shall then be reduced to zero psig (normal atmospheric pressure). The periscope shall then be flushed with nitrogen to the requirements of 3.7. The interior atmosphere of the periscope shall be periodically sampled through a dew point meter until the atmosphere shows a dew point at least as low as -25°F. When this atmosphere is attained witnin the periscope, the periscope shall be sealed with an internal pressure of nitrogen at a pressure of between 0.25 and 0.50 psig at room temperature.

4.6.5 Cleanliness.- The optical surfaces of the periscope assembly shall be inspected for cleanliness in accordance with MIL-O-13830. The periscope assembly shall conform to the cleanliness requirements of 3.8.

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4.6.6 Resolution.- Position the periscope on a holding device capable of accepting the locating bushing of the periscope assembly. With the filters of the periscope in clear position, perform this test utilizing an observation telescope with a magnification of at least 3 power (3X) and a resolving power chart on a wall target. The test methods and procedures shall be in compliance with the "Resolution Test" specified in MIL-O-13830 and the resolving power chart shall contain four line sets as shown in Figure 1 therein. The observation telescope shall be positioned at each eyepiece assembly of the periscope. The focus of the observation telescope and eyepiece shall be established at one best focus setting for viewing of the detectable line structure and proper line count of the resolution target. Resolution of each optical system shall meet the requirements specified in 3.9.4.

4.6.7 Orientation.- Perform the following tests with the periscope assembly positioned on special testing equipment conforming to 4.5.2.3.1.

4.6.7.1 Scales and level vial.- The angle of site scale, micrometer knob scale and level vial bubble of the periscope shall meet the requirements specified in 3.9.1. Record the position of the level vial and scales.

4.6.7.2 Level line.- Observe through the right eyepiece of the periscope. The line of sight through the geometric axis of the periscope reticle pattern shall meet the requirements specified in 3.9.2.

4.6.7.3 Collimation.- Perform the test with the aid of the viewing double collimator positioned at the eyepiece assemblies of the periscope. With the filters of the periscope in clear position, set the diopter scales of the periscope to indicate zero and the interpupillary distance knob (IPD) positioned at 66 millimeters (MM). View through the right eyepiece of the viewing double collimator. Adjust the collimator in elevation or depression to establish coincidence of the intersecting point of the zero vertical and horizontal reticle lines of the collimator with respect to the intersecting point of the zero vertical and horizontal target image lines. Observe through the left eyepiece of the collimator, the left zero vertical target image line shall appear on the side of the zero vertical reticle line of the collimator which represents divergence of the target images. Measure the distance from the right target image to the left target image in elevation (dipvergence). The emerging rays from the two eyepieces shall be within the tolerance specified in 3.9.3. Record the divergence and dipvergence readings. Repeat the above procedure with the IPD knob positioned at 60MM and 72MM setting. Record the divergence and dipvergence readings. The requirements specified in 3.9.3 shall be met.

4.6.7.4 Parallax.- Perform this test with the right diopter knob scale of the periscope set for best focus of the periscope reticle. Parallax requirements shall be met when the right target image is adjusted as specified in 3.9.5. Parallax will be recognized by any apparent displacement of the periscope reticle center lines with respect to the center target image lines when the observer's head is moved from side to side or up and down. Displacement between the periscope reticle and target image shall not exceed the tolerance specified in 3.9.5.

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4.6.7.5 Magnification.- The right and left optical systems of the periscope shall be inspected for magnification utilizing a dynameter and a diaphragm with a known diameter. Position the diaphragm at the entrance window of the periscope incident to the right objective. Set the dynameter at the right eyepiece. While viewing through the dynameter, adjust the dynameter for best focus of the diaphragm image. Measure the diameter of this image utilizing the millimeter reticle scale of the dynameter. Divide the known diameter by the diaphragm image diameter as read on the dynameter reticle scale. Record this value. Repeat the magnification test procedure above for the left optical system. Record this value. Any difference between the two recorded values shall not exceed the tolerance specified in 3.9.6.

4.6.7.6 Filters.- Perform this test in sequence with the test in 4.6.7.5. Position each filter knob to read (c) and view through each eyepiece assembly of the periscope. The clear filters shall be in the field of view with no overlap of adjacent filters. Introduction of the remaining filters into each optical system shall meet the requirements specified in 3.9.7 with respect to their filter selector knobs.

4.6.7.7 Reticle accuracy.- This test shall be conducted in accordance with test for reticle scale spacings of MIL-O-13830. The total angular subtense of the reticle graduations shall meet the requirements specified in 3.9.8 as measured on the target image.

4.6.7.8 Image tilt.- The right and left optical systems of the periscope assembly shall be inspected for image tilt utilizing a calibrated observation telescope capable of measuring image tilt over a field of view of 80 mils and an appropriate reticle pattern to determine image tilt accuracies specified in 3.9.9. The vertical reticle line of the observation telescope shall be aligned plumb when the test position on a qualified level surface at each eyepiece of the periscope assembly. Set the observation telescope at the right eyepiece of the periscope assembly and sight through the observation telescope to establish one extremity of the observation telescope vertical reticle line coincident to a vertical target image line. The vertical reticle line of the target image with respect to the vertical reticle line of the observation telescope shall be within 1.6 mils at the edge of the field when measured over a field of view of 80 mils. Repeat the above procedure with the observation telescope positioned at the left eyepiece of the periscope assembly. The images formed by the two optical systems of the periscope assembly shall meet the requirements specified in 3.9.9.

4.6.7.9 Parallelism of reticle and image.- Perform this test while observing through the right eyepiece of the periscope with the unaided eye. One extremity of the periscope vertical reticle line shall be coincident to a vertical target image line. The reticle line shall be parallel to the vertical target image line within 0.8 mils at the edge of the field when measured over a field of view of 80 mils.

4.6.7.10 Eyepiece focus.- A calibrated dioptometer with a magnification of at least 3 power (3X) shall be utilized for measuring the accuracy

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of the right and left zero diopter settings of the diopter scales Set the dioptometer objective scale to zero and adjust the dioptometer eyepiece for best focus of the dioptometer reticle. Set the right diopter knob of the periscope eyepiece assembly to indicate zero diopter. Position the dioptometer at the eyepiece of the periscope. Observe through the dioptometer, obtain best focus of the right target image by adjustment of the periscope eyepiece diopter knob. Repeat the above test procedure for best focus of the left target image utilizing the left eyepiece assembly. Each target image shall be in best focus within the tolerance specified in 3.9.11, as measured on the diopter knob scales of the periscope. Rotate each diopter knob from the established best focus of the target images to its clockwise stop, note the diopter setting on the scales, then rotate each diopter knob to its counterclockwise stop, note diopter setting on the scales. Each diopter movement shall meet the requirements specified in 3.9.11.

4.6.7.11 Reticle focus.- Perform this test while observing through the right eyepiece of the periscope with the unaided eye and obtain best focus of the periscope reticle by adjustment of the periscope eyepiece diopter knob. Any difference in focus between the horizontal and vertical lines of the reticle image shall not exceed the tolerance specified in 3.9.12 as measured on the diopter knob scale.

4.6.7.12 Interpupillary adjustment.- The interpupillary distance between each eyepiece shall be measured by means of the exit pupil diameters (edge to edge) utilizing a millimeter scale. The interpupillary scale shall meet the requirements of 3.9.13.

4.6.7.13 Azimuth displacement.- Perform this test with the angle of site mechanism positioned as specified in 3.9.17. Observe through the right eyepiece of the periscope and establish coincidence of the periscope vertical reticle line adjacent to the 4 mil circle with respect to the intersecting point of the zero horizontal and vertical target image lines utilizing the periscope elevation knob. Rotate the elevation knob with the angular velocity specified in 3.9.17 to elevate the line of sight of the periscope to 300 mils. Any azimuth displacement, between the vertical reticle line of the periscope with respect to the target image line due to reversing the travel of the elevation knob to reestablish zero elevation shall not exceed the tolerance specified in 3.9.17 as measured on the target image line.

4.6.8 Performance - periscope assembly.- Perform the following tests utilizing special testing equipment conforming to 4.5.2.3.2.

4.6.8.1 Elevation excursion.- While observing through the right eyepiece of the periscope assembly, utilize the elevation knob to depress the line of sight of the periscope to -320 mils. The "0" horizontal line of the periscope reticle shall be coincident with respect to the -320 mil horizontal target image line. Rotate the elevation knob to elevate the line of sight to +320 mils above the "0" (horizontal line of sight). The line of sight of the periscope shall meet the requirements specified in 3.9.14 as measured on the target image.

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4.6.8.2 Chatter in elevation or depression movement.- Perform this test with the elevation fixture positioned at the following depression and elevation settings in mils -300, -200, -100, "0" (horizontal line of sight), +100, +200 and +300 mils. While sighting through the right eyepiece of the periscope, utilize the elevation knob of the periscope to establish coincidence of the "0" horizontal line of the periscope reticle with respect to horizontal target image lines corresponding to the depression and elevation settings introduced on the elevation fixture. The vertical displacement readings shall be made at 100 mils increments. As each 100 mil increment is established on the elevation test fixture, rotate the elevation knob of the periscope to establish coincidence of the "0" horizontal periscope reticle line with respect to the horizontal target image line. Any vertical displacement, between the target image lines with respect to the horizontal reticle line of the periscope due to chatter in the elevation movement to establish coincidence, shall not exceed the tolerance specified in 3.9.15 as measured on the target image. Repeat the above procedure for depression movement, except the "0" horizontal line of the periscope reticle shall be in coincidence with respect to the +300 mil horizontal target image line. Chatter in elevation or depression movement shall be within the tolerance specified in 3.9.15.

4.6.8.3 Plumb travel.- Perform this test with the elevation fixture positioned at -320 mils. Observe through the right eyepiece of the periscope and depress the line of sight of the periscope to -320 mils below the "0" (horizontal line of sight) of the target image utilizing the periscope elevation knob. The vertical line adjacent to the 4 mil circle of the periscope reticle shall be in coincidence with respect to the vertical target image line. While rotating the elevation knob of the periscope, simultaneously rotate the elevation fixture to elevate the line of sight of the periscope to +320 mils above the "0" (horizontal line of sight) of the target image. During this excursion the vertical line of the periscope reticle shall be in coincidence with respect to the verticle target image line within the tolerance specified in 3.9.16. The total deviation between any two error readings shall not exceed the tolerance specified in 3.9.16.

4.6.8.4 Angle of site error.- Perform this test with the elevation fixture positioned at the following depression and elevation settings in mils -300, -200, -100, "0" (horizontal line of sight), +100, +200 and +300 mils. While sighting through the right eyepiece of the periscope, utilize the elevation knob of the periscope to establish coincidence of the zero horizontal line of the periscope reticle with respect to target images corresponding to the depression and elevation settings introduced on the elevation fixture. The angle of site readings shall be made at 100 mil increments. As each 100 mil increment is established on the elevation test fixture and periscope assembly, rotate the angle of site mechanism by means of the micrometer knob to position the level vial and micrometer knob scale to the established position of 4.6.7.1 without overtravel. Record the angle of site scale reading with respect to its index as determined by the level vial and micrometer knob scale reading at each 100 mil increment position throughout 600 mils of angle of site excursion. Each recorded angle of site scale reading shall meet the requirements specified in 3.9.16 as measured on the micrometer knob scale.

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NOTE: During this operation (4.6.8.4) if overtravel occurs, the test shall be nullified and the procedure for angle of site accuracy shall be repeated.

4.6.8.5 Backlash angle of site.- Perform this test in conjunction with the test in 4.6.8.4. With backlash excluded for the direction of travel of the angle of site mechanism, position the level vial and micrometer knob scale to the recorded values established at each 100 mil increment in 4.6.8.4 without overtravel. Rotate the micrometer knob one complete revolution in the same direction of travel, then reverse the direction of travel of the knob with backlash included to return the level vial to its recorded value without overtravel. Record the backlash value of the angle of site scale with respect to its index at each 100 mil increment position throughout 600 mils of excursion as determined by the micrometer knob scale. The values in backlash of the angle of site mechanism shall not exceed the tolerance specified in 3.9.19 as measured on the micrometer knob scale.

NOTE: During this operation (4.6.8.5) if overtravel occurs, the test shall be nullified and the procedure for backlash shall be repeated.

4.6.9 Illumination.- The testing equipment utilized in this test shall conform to the requirements specified in 3.9.20. With the conditions specified in 3.9.20 applied to the periscope, the illuminated reticle shall be clearly defined when visually examined. Should the brightness of a reticle defect be greater than the etched lines, the defect shall be cause for rejection.

4.6.10 Operability.- Standard torque measuring equipment capable of measuring the torque values specified in 3.9.21, Table I shall be utilized for these tests. Appropriate adapters to accomodate the components under tests shall be used with the torque measuring equipment. Torque shall be measured while rotating the components through their full range of travel. The measured torque values shall meet the requirements specified in 3.9.21, Table I at the temperatures indicated. Subsequent to the torque tests and while still at the temperature extreme the periscope shall be subjected to the applicable functional operability tests to determine conformance to the requirements of 3.9.21, Table I at the temperature indicated.

4.6.11 Workmanship.- The periscope assembly shall be subjected to a visual examination and shall meet the requirements specified in 3.4.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- The Battery Command M65 shall be placed in Chest, Packing, M39A1, Part Number 7694908, Federal Stock Number 6650-769-4908. Packaging of the periscope and the case shall be in accordance with Packaging Data Sheet 8268671. The level of protection shall be as specified in the procurement document (see 6.2 and 6.3).

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6. NOTES

6.1 Intended use.- The Battery Command Periscope, M65 is a binocular periscope type of instrument of optical mechanical assembly having 10 power magnification with a six degree field of view. It is primarily intended to observe enemy targets and measure angles in azimuth and elevation.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of applicable levels of preservation, packaging and packing
(c) Applicable stock number
(d) Applicable packaging data sheet (see 5.1)
(e) Availability and quantity required of Chest, Packing M39A1 (see 6.3)

6.3 Chest, Packing M39A1.- The contracting officer shall establish the availability of Packing Chest M39A1 to the suppliers of Battery Command Periscope M65. The chest shall be fabricated to drawing F7094093.

Custodian:

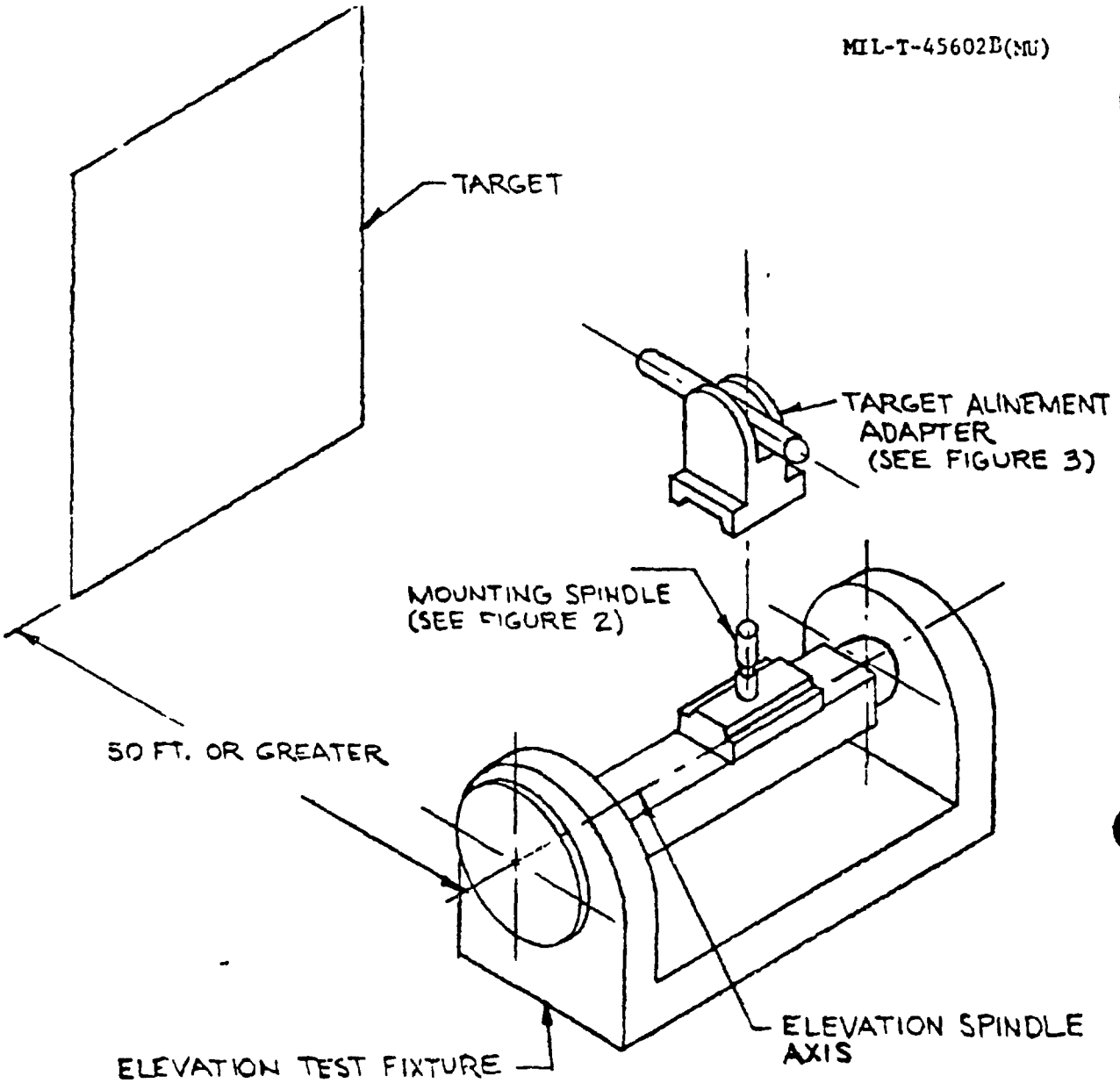
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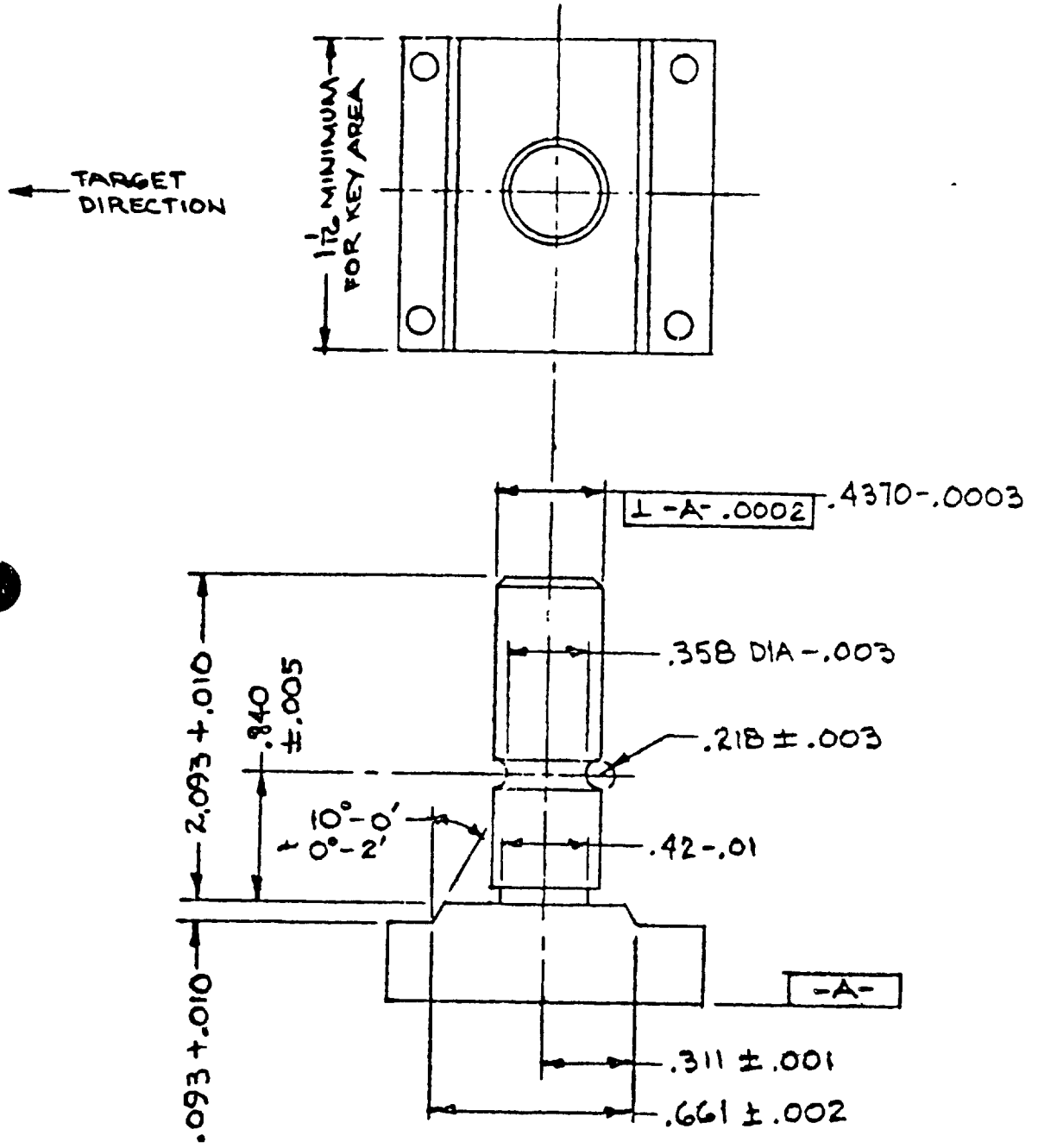


ELEVATION TESTS SET-UP

FIGURE 1

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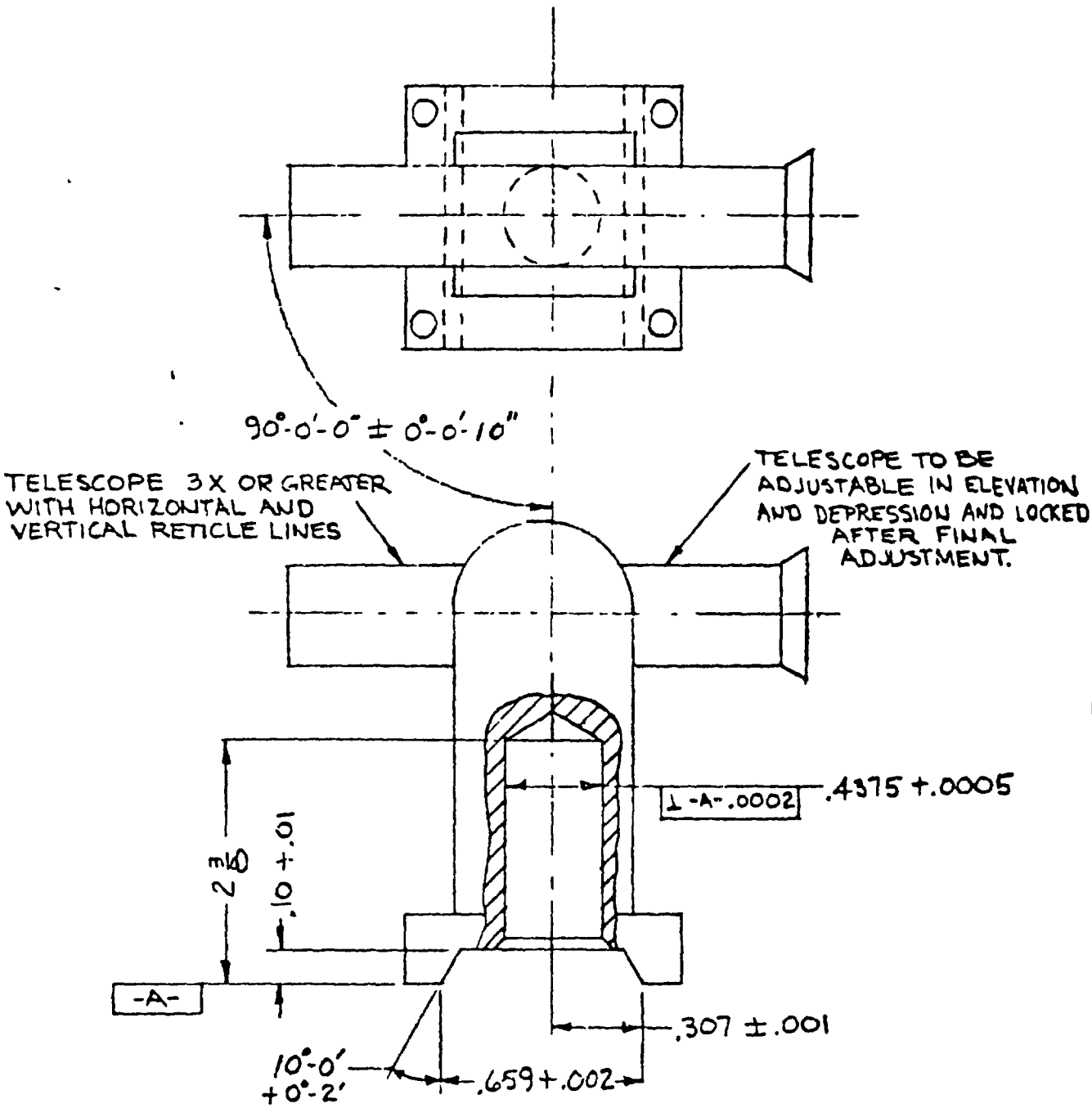
NOTES:
1. THE SIZE IS OPTIONAL ON UNDIMENSIONED FEATURES.
2. FIGURE IS NOT TO SCALE.



TELESCOPE MOUNTING SPINDLE

FIGURE 2

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NOTES:
1. THE SIZE AND SHAPE IS OPTIONAL ON UNDIMENSIONED FEATURES.
2. FIGURE IS NOT TO SCALE.

TARGET ALINEMENT ADAPTER

FIGURE 3

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R355

[illegible] either Government or contractor involved in the [illegible] for ultimate use by the Department of Defense. This sheet is [illegible] the use of this specification which will insure that suitable products are procured at the minimum overall cost in the least time and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

a. NAME OF ORGANIZATION

b. ADDRESS

c. CONTRACT NUMBER

d. TYPE OF ORGANIZATION
DIRECT GOVERNMENT CONTRACT SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

a. GIVE PARAGRAPH NUMBER AND WORDING

b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES NO (If "yes" in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)

5. SUBMITTED BY (Printed or typed name and activity. Optional)

DATE

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